CLAIMS

What is claimed is:

1. A method of identifying a subject having a cancer who is likely to benefit from a therapy, comprising:

acquiring a value for the level or activity of CD4+ immune effector cells, (*e.g.*, CD4+ T cells), relative to the level or activity of CD8+ immune effector cells, (*e.g.*, CD8+ T cells,) in the subject, (*e.g.*, in a sample from the subject);, wherein

wherein (a) a value that is greater than or equal to a reference value identifies the subject as one who is likely to benefit from the therapy, and

wherein the therapy comprises a targeted therapy in combination with an immuno-oncology therapy; or

(b) a value that is less than a reference value identifies the subject as one who is likely to benefit from the therapy, and

wherein the therapy comprises a targeted therapy, *e.g.*, without an immuno-oncology therapy.

2. The method of claim 1, wherein the subject identified as having a value that is greater than or equal to the reference value is likely to have an increased benefit from the therapy, as compared to a therapy comprising a targeted therapy without an immuno-oncology therapy; or 93. The method of claim 92, wherein the subject identified as having a value that is less than the reference value is not likely to have a substantially increased benefit from a therapy comprising the targeted therapy in combination with an immuno-oncology therapy.

33. A method of selecting a therapy for a subject having a cancer, the method comprising:

acquiring a value for the level or activity of CD4+ immune effector cells, (*e.g.*, CD4+ T cells), relative to the level or activity of CD8+ immune effector cells, (*e.g.*, CD8+ T cells), in the subject, (*e.g.*, in a sample from the subject);, and

(a) if the value is greater than or equal to a reference value, selecting a therapy comprising a targeted therapy in combination with an immuno-oncology therapy for the subject; or

(a)(b) if the value is less than a reference value, selecting a therapy comprising a targeted therapy, *e.g.*, without an immuno-oncology therapy, for the subject.

4177. A method of stratifying subjects having a cancer into a first group and a second group, comprising:

acquiring a value for the level or activity of CD4+ immune effector cells, (*e.g.*, CD4+ T cells,) relative to the level or activity of CD8+ immune effector cells, (*e.g.*, CD8+ T cells,) in the subject, (*e.g.*, in a sample from the subject),

assigning a subject who has a value that is less than a reference value to the first group who is likely to benefit from a therapy comprising a targeted therapy, (*e.g.*, without an immuno-oncology therapy), and

assigning a subject who has a value that is greater than or equal to a reference value to the second group which is likely to benefit from a therapy comprising a targeted therapy in combination with an immuno-oncology therapy.

5178. A method of stratifying subjects having a cancer into a first group and a second group for selecting a therapy, comprising:

acquiring a value for the level or activity of CD4+ immune effector cells, (*e.g.*, CD4+ T cells,) relative to the level or activity of CD8+ immune effector cells, (*e.g.*, CD8+ T cells,) in the subject, (*e.g.*, in a sample from the subject),

wherein a value that is less than a reference value identifies a subject as a member of the first group which is likely to benefit from a therapy comprising a targeted therapy, (*e.g.*, without an immuno-oncology therapy), and

wherein a value that is greater than or equal to a reference value identifies a subject as a member of the second group which is likely to benefit from a therapy comprising a targeted therapy in combination with an immunotherapy.

64. The method of any ~~one of~~preceding claim~~s 1-3~~, further comprising:

administering, (*e.g.*, initiating administering or continuing administering,) an effective amount of the therapy to the subject;

~~5. The method of any one of claims 1-3, further comprising~~ administering an altered dosing regimen of the therapy, (*e.g.*, a dosing regimen with a higher dose and/or more frequent administration than a reference dosing regimen,) to the subject;

~~6. The method of any one of claims 1-3, further comprising~~ discontinuing administration of a different therapy to the subject;

~~7. The method of any one of claims 1-3, further comprising~~ administering an additional therapy to the subject; or

~~8. The method of claim 7, further comprising~~

administering a pretreatment to the subject, wherein the pretreatment increases the value for the level or activity of CD4+ immune effector cells, (*e.g.*, CD4+ T cells,) relative to the level or activity of CD8+ immune effector cells, (*e.g.*, CD8+ T cells,) in the subject, (*e.g.*, in a sample from the subject).

728. A therapy comprising a targeted therapy or comprising a targeted therapy in combination with an immuno-oncology therapy for use in a method of treating a subject having a cancer, wherein the method comprises(ing):

responsive to a value for the level or activity of CD4+ immune effector cells, (*e.g.*, CD4+ T cells,) relative to the level or activity of CD8+ immune effector cells, (*e.g.*, CD8+ T cells,) in the subject, (*e.g.*, in a sample from the subject),

administering, (*e.g.*, initiating administering or continuing administering,) an effective amount of the therapy comprising the targeted therapy, (*e.g.*, without an immuno-oncology therapy,) to a subject having a value that is less than a reference value; or

administering, (*e.g.*, initiating administering or continuing administering,) an effective amount of the therapy comprising the targeted therapy in combination with the immuno-oncology therapy to a subject having a value that is greater than or equal to a reference value.

8289. A therapy comprising a targeted therapy or comprising a targeted therapy in combination with an immuno-oncology therapy for use in a method of treating a subject having a cancer, wherein the method comprises(ing):

acquiring a value for the level or activity of CD4+ immune effector cells, (*e.g.*, CD4+ T cells,) relative to the level or activity of CD8+ immune effector cells, (*e.g.*, CD8+ T cells,) in the subject, (*e.g.*, in a sample from the subject); and

administering, (*e.g.*, initiating administering or continuing administering,) an effective amount of the therapy comprising the targeted therapy, (*e.g.*, without an immuno-oncology therapy,) to a subject having a value that is less than a reference value; or

administering, (*e.g.*, initiating administering or continuing administering,) an effective amount of the therapy comprising the targeted therapy in combination with the immuno-oncology therapy to a subject having a value that is greater than or equal to a reference value.

9290. A therapy comprising a targeted therapy or comprising a targeted therapy in combination with an immuno-oncology therapy for use in a method of treating a subject having a cancer, wherein the method comprises(ing):

administering, (*e.g.*, initiating administering or continuing administering,) an effective amount of the therapy comprising the targeted therapy, (*e.g.*, without an immuno-oncology therapy,) to the subject;

responsive to a value for the level or activity of CD4+ immune effector cells, (*e.g.*, CD4+ T cells,) relative to the level or activity of CD8+ immune effector cells, (*e.g.*, CD8+ T cells,) in the subject, (*e.g.*, in a sample from the subject),

administering, (*e.g.*, initiating administering or continuing administering,) the therapy comprising the targeted therapy, (*e.g.*, without an immuno-oncology therapy,) for a subject having a value that is less than a reference value, or

administering, (*e.g.*, initiating administering or continuing administering,) the therapy comprising the targeted therapy in combination with the immuno-oncology therapy for a subject having a value that is greater than or equal to a reference value.

10291. A therapy comprising a targeted therapy or comprising a targeted therapy in combination with an immuno-oncology therapy for use in a method of treating a subject having a cancer, wherein the method comprises~~ing~~ administering to the subject an effective amount of the therapy comprising:

(a) the targeted therapy, (*e.g.*, without an immuno-oncology therapy,), wherein prior to the administration, a value for the level or activity of CD4+ immune effector cells, (*e.g.*, CD4+ T cells,) relative to the level or activity of CD8+ immune effector cells, (*e.g.*, CD8+ T cells,) in the subject, (*e.g.*, in a sample from the subject,), that is less than a reference value, has been determined; or

(b) the targeted therapy in combination with the immuno-oncology therapy, wherein prior to the administration, a value for the level or activity of CD4+ immune effector cells, (*e.g.*, CD4+ T cells,) relative to the level or activity of CD8+ immune effector cells, (*e.g.*, CD8+ T cells,) in the subject, (*e.g.*, in a sample from the subject,), that is greater than or equal to a reference value, has been determined.

11292. A therapy comprising a targeted therapy or comprising a targeted therapy in combination with an immuno-oncology therapy for use in a method of treating a subject having a cancer, wherein the method comprises~~ing~~ administering to the subject an effective amount of the therapy comprising:

(a) the targeted therapy, (*e.g.*, without an immuno-oncology therapy,) to the subject, wherein the subject is, or has been, characterized as having a value for the level or activity of CD4+ immune effector cells, (*e.g.*, CD4+ T cells,) relative to the level or activity of CD8+ immune effector cells, (*e.g.*, CD8+ T cells,) in the subject, (*e.g.*, in a sample from the subject), that is less than a reference value; or

(b) the targeted therapy in combination with the immuno-oncology therapy, wherein the subject is, or has been, characterized as having a value for the level or activity of CD4+ immune effector cells, (*e.g.*, CD4+ T cells,) relative to the level or activity of CD8+ immune effector cells, (*e.g.*, CD8+ T cells,) in the subject, (*e.g.*, in a sample from the subject), that is greater than or equal to a reference value.

9. A method of treating a subject having a cancer, comprising:

responsive to a value for the level or activity of CD4+ immune effector cells (*e.g.*, CD4+ T cells) relative to the level or activity of CD8+ immune effector cells (*e.g.*, CD8+ T cells) in the subject (*e.g.*, in a sample from the subject), that is greater than or equal to a reference value,

administering (*e.g.*, initiating administering or continuing administering) an effective amount of a therapy comprising a targeted therapy in combination with an immuno-oncology therapy to the subject,

thereby treating the subject having the cancer.

10. A method of treating a subject having a cancer, comprising:

acquiring a value for the level or activity of CD4+ immune effector cells (*e.g.*, CD4+ T cells) relative to the level or activity of CD8+ immune effector cells (*e.g.*, CD8+ T cells) in the subject (*e.g.*, in a sample from the subject),

if the value is greater than or equal to a reference value, administering (*e.g.*, initiating administering or continuing administering) an effective amount of a therapy comprising a targeted therapy in combination with an immuno-oncology therapy to the subject,

thereby treating the subject having the cancer.

11. A method of treating a subject having a cancer, comprising:

administering (*e.g.*, initiating administering or continuing administering) an effective amount of a targeted therapy to the subject;

responsive to a value for the level or activity of CD4+ immune effector cells (*e.g.*, CD4+ T cells) relative to the level or activity of CD8+ immune effector cells (*e.g.*, CD8+ T cells) in the subject (*e.g.*, in a sample from the subject), that is greater than or equal to a reference value,

administering (*e.g.*, initiating administering or continuing administering) an effective amount of an immuno-oncology therapy to the subject,

thereby treating the subject having the cancer.

12. A method of treating a subject having a cancer, comprising:

administering (*e.g.*, initiating administering or continuing administering) an effective amount of a therapy comprising a targeted therapy in combination with an immuno-oncology therapy to the subject,

wherein prior to the administration, a value for the level or activity of CD4+ immune effector cells (*e.g.*, CD4+ T cells) relative to the level or activity of CD8+ immune effector cells (*e.g.*, CD8+ T cells) in the subject (*e.g.*, in a sample from the subject), that is greater than or equal to a reference value, has been determined,

thereby treating the subject having the cancer.

13. A method of treating a subject having a cancer, comprising:

administering (*e.g.*, initiating administering or continuing administering) an effective amount of a therapy comprising a targeted therapy in combination with an immuno-oncology therapy to the subject,

wherein the subject is characterized as having a value for the level or activity of CD4+ immune effector cells (*e.g.*, CD4+ T cells) relative to the level or activity of CD8+ immune effector cells (*e.g.*, CD8+ T cells) in the subject (*e.g.*, in a sample from the subject), that is greater than or equal to a reference value,

thereby treating the subject having the cancer.

14. The method of any of claims 1-13, wherein the value for the level or activity of CD4+ immune effector cells (*e.g.*, CD4+ T cells) relative to the level or activity of CD8+ immune effector cells (*e.g.*, CD8+ T cells) comprises a ratio of the amount of CD4+ immune effector cells (*e.g.*, CD4+ T cells) to the amount of CD8+ immune effector cells (*e.g.*, CD8+ T cells), *e.g.*, as measured by an assay disclosed herein, *e.g.*, flow cytometry immunophenotyping.

15. The method of claim 14, wherein the value is greater than or equal to 2 (e.g., 2.01).

16. The method of claim 15, wherein the value is greater than or equal to 2.5, 3, 3.5, 4, 4.5, 5, 5.5, 6, 6.5, 7, 7.5, 8, 8.5, 9, 9.5, or 10.

17. The method of claim 14, wherein the value is greater than or equal to 3.3 (e.g., 3.34).

18. The method of claim 17, wherein the value is greater than or equal to 3.5, 4, 4.5, 5, 5.5, 6, 6.5, 7, 7.5, 8, 8.5, 9, 9.5, or 10.

19. The method of any one of claims 1-18, wherein acquiring the value comprises determining the level or activity of CD4+ immune effector cells (*e.g.*, CD4+ T cells) relative to the level or activity of CD8+ immune effector cells (*e.g.*, CD8+ T cells) in the subject, *e.g.*, in a sample from the subject.

20. The method of claim 19, wherein the sample from the subject comprises a blood sample (*e.g.*, a peripheral blood sample, *e.g.*, comprising peripheral blood mononuclear cells (PBMCs)) or a tumor sample.

21. The method of claim 19 or 20, wherein the value is acquired before administration of the therapy is initiated (*e.g.*, is a baseline value).

22. The method of claim 19 or 20, wherein the value is acquired after administration of the therapy is initiated.

23. The method of claim 22, wherein the value is acquired 1, 2, 3, 4, 8, 10, 12, 20, 30, 40 weeks or more or 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 12, 14, 16, 18, 20 months, or more after administration of the therapy is initiated.

24. The method of any one of claims 1-23, further comprising acquiring a value for the level or activity of CD8+ tumor infiltrating lymphocytes (TILs), *e.g.*, tumors having CD8+ TILs inflamed phenotype, in the subject (*e.g.*, in a sample from the subject).

25. The method of claim 24, wherein an increase in the value for the level or activity of CD8+ TILs, as compared to a reference value, further identifies the subject as one who is likely to benefit from the therapy.

26. The method of any one of claims 1-25, further comprising acquiring a value for tumor mutation burden (TMB), in the subject (*e.g.*, in a sample from the subject).

27. The method of claim 26, wherein an increased value for TMB, as compared to a reference value, further identifies the subject as one who is likely to benefit from the therapy.

28. The method of claim 27, wherein the value for TMB is greater than or equal to 10 mut/Mb, *e.g.*, greater than or equal to 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25 mut/Mb, or more.

29. The method of any one of claims 1-28, further comprising acquiring a value for the level and/or activity of PD-L1 in the subject (*e.g.*, in a sample from the subject).

30. The method of claim 29, wherein a decreased value for the level and/or activity of PD-L1, as compared to a reference value, *e.g.*, together with an increased value for TMB, as compared to a reference value, further identifies the subject as one who is likely to benefit from the therapy.

31. The method of claim 30, wherein the cancer has low or no detectable expression of PD-L1.

32. The method of any one of claims 1-31 further comprising acquiring a value for circulating tumor DNA (ctDNA) in the subject (*e.g.*, in a sample from the subject).

33. The method of claim 32, wherein an increased value for ctDNA, as compared to a reference value, further identifies the subject as one who is likely to benefit from the therapy.

34. The method of any one of claims 1-33, wherein a subject who is likely to benefit from, or is likely to have an increased benefit from, the therapy has an improved progression-free survival (PFS), duration of objective response (DOR), and/or overall survival (OS), compared to a subject who is unlikely to benefit from the therapy, or is unlikely to have an increased benefit from the therapy.

35. The method of any one of claims 1-33, wherein a subject who is likely to benefit from, or is likely to have an increased benefit from, the therapy has an improved PFS, DOR, and/or OS, compared to a subject who has not received the therapy, or has only received targeted therapy but not an immuno-oncology therapy.

36. The method of claim 34 or 35, wherein the PFS, DOR, and/or OS is improved by 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 18, 24, 30, 36, 42, 48, 54, 60 months or more.

37. The method of claim 34 or 35, wherein the PFS, DOR, and/or OS is improved by 10%, 20%, 30%, 40%, 50%, 60%, 70%, 80%, or 90%, or 1, 2, 3, 4, 5, 6, 7, 8, 9, 10-fold or more.

38. The method of any one of claims 1-37, wherein the subject has been treated, or is being treated, with a targeted therapy.

39. The method of any one of claims 1-37, wherein the subject has not been treated, or is not being treated, with a targeted therapy.

40. The method of any one of claims 1-37, wherein the subject has received, or is receiving, with an immuno-oncology therapy.

41. The method of any one of claims 1-37, wherein the subject has not received, or is not receiving, with an immuno-oncology therapy.

42. The method of claim any one of claims 1-37, wherein the subject has received, or is receiving, with a therapy comprising a targeted therapy in combination with an immuno-oncology therapy.

43. The method of claim any one of claims 1-37, wherein the subject has not received, or is not receiving, with a therapy comprising a targeted therapy in combination with an immuno-oncology therapy.

44. The method of any one of claims 1-37, wherein the subject has received, or is receiving, with a targeted therapy, and the cancer has relapsed.

45. The method of any one of claims 1-37, wherein the subject is, or has been identified as, a non-responder to a targeted therapy.

46. The method of any one of claims 1-37, wherein the subject is, or has been identified as, a partial responder to a targeted therapy.

47. The method of any one of claims 1-46, wherein the targeted therapy comprises an agent targeting BRAF and/or an agent targeting MEK.

48. The method of claim 47, wherein the targeted therapy comprises an agent targeting BRAF or an agent targeting MEK.

49. The method of claim 47 or 48, wherein the agent targeting BRAF is a BRAF inhibitor.

50. The method of any one of claims 47-49, wherein the agent targeting BRAF inhibits wild-type BRAF and/or BRAF having a V600 mutation (*e.g.*, a V600E mutation or a V600K mutation).

51. The method of any one of claims 47-50, wherein the agent targeting BRAF is dabrafenib, vemurafenib, encorafenib, ABM-1310, ARQ-736, ASN003, BGB-283, BGB-3245, CEP-32496, GDC-0879, LUT014, PLX4720, PLX8394, RO5212054, or a pharmaceutically acceptable salt thereof.

52. The method of claim 51, wherein the agent targeting BRAF is dabrafenib.

53. The method of claim 51, wherein the agent targeting BRAF is vemurafenib.

54. The method of any one of claims 47-52, wherein the agent targeting BRAF (*e.g.* dabrafenib) is administered (*e.g.*, orally) at a dose between 25 mg and 300 mg (*e.g.*, between 50 mg and 250 mg or between 100 mg and 200 mg, *e.g.*, 150 mg), *e.g.*, twice a day.

55. The method of any one of claims 47-54, wherein the agent targeting MEK is a MEK inhibitor.

56. The method of claim 47-55, wherein the agent targeting MEK is trametinib, cobimetinib, binimetinib, mirdametinib, pimasertib, refametinib, selumetinib, AS703988, AZD 8330, BI 847325, BIX 02188, BIX 02189, CI-1040, CS3006, E6201, FCN-159, G-38963, GDC-0623, HL-085, PD 98059, RO4987655, RO5126766, SHR 7390, TAK-733, U0126, WX-554, or a pharmaceutically acceptable salt thereof.

57. The method of claim 56, wherein the agent targeting MEK is trametinib.

58. The method of claim 56, wherein the agent targeting MEK is cobimetinib.

59. The method of any one of claims 47-57, wherein the agent targeting MEK (*e.g.*, trametinib) is administered (*e.g.*, orally) at a dose between 0.1 mg and 5 mg (*e.g.*, between 0.5 mg and 4 mg or between 1 mg and 3 mg, *e.g.*, at a dose of 2 mg), *e.g.*, once a day.

60. The method of any one of claims 47, 49-52, 54-57, or 59, wherein the agent targeting BRAF is dabrafenib and the agent targeting MEK is trametinib.

61. The method of claim 47, 49-51, 53-56, 58, or 59, wherein the agent targeting BRAF is vemurafenib and the agent targeting MEK is cobimetinib.

62. The method of any one of claims 1-61, wherein the immuno-oncology therapy comprises a PD-1 or PD-L1 binding antagonist.

63. The method of any one of claims 1-62, wherein the immuno-oncology therapy comprises an PD-1 inhibitor.

64. The method of any one of claims 1-63, wherein the immuno-oncology therapy comprises an anti-PD-1 antibody molecule.

65. The method of claim 64, wherein the anti-PD-1 antibody molecule comprises:

a heavy chain variable region (VH) comprising a VHCDR1 amino acid sequence of SEQ ID NO: 501, a VHCDR2 amino acid sequence of SEQ ID NO: 502, and a VHCDR3 amino acid sequence of SEQ ID NO: 503; and

a light chain variable region (VL) comprising a VLCDR1 amino acid sequence of SEQ ID NO: 510, a VLCDR2 amino acid sequence of SEQ ID NO: 511, and a VLCDR3 amino acid sequence of SEQ ID NO: 512, each disclosed in Table 1.

66. The method of claim 65, wherein the anti-PD-1 antibody molecule comprises a VH comprising the amino acid sequence of SEQ ID NO: 506, or an amino acid sequence at least 85%, 90%, 95%, or 99% identical or higher to SEQ ID NO: 506.

67. The method of claim 65, wherein the anti-PD-1 antibody molecule comprises a VL comprising the amino acid sequence of SEQ ID NO: 520, or an amino acid sequence at least 85%, 90%, 95%, or 99% identical or higher to SEQ ID NO: 520.

68. The method of claim 65, wherein the anti-PD-1 antibody molecule comprises a VL comprising the amino acid sequence of SEQ ID NO: 516, or an amino acid sequence at least 85%, 90%, 95%, or 99% identical or higher to SEQ ID NO: 516.

69. The method of claim 65, wherein the anti-PD-1 antibody molecule comprises a VH comprising the amino acid sequence of SEQ ID NO: 506 and a VL comprising the amino acid sequence of SEQ ID NO: 520.

70. The method of claim 65, wherein the anti-PD-1 antibody molecule comprises a VH comprising the amino acid sequence of SEQ ID NO: 506 and a VL comprising the amino acid sequence of SEQ ID NO: 516.

71. The method of claim 65, wherein the anti-PD-1 antibody molecule comprises a heavy chain comprising the amino acid sequence of SEQ ID NO: 508, or an amino acid sequence at least 85%, 90%, 95%, or 99% identical or higher to SEQ ID NO: 508.

72. The method of claim 65, wherein the anti-PD-1 antibody molecule comprises a light chain comprising the amino acid sequence of SEQ ID NO: 522, or an amino acid sequence at least 85%, 90%, 95%, or 99% identical or higher to SEQ ID NO: 522.

73. The method of claim 65, wherein the anti-PD-1 antibody molecule comprises a light chain comprising the amino acid sequence of SEQ ID NO: 518, or an amino acid sequence at least 85%, 90%, 95%, or 99% identical or higher to SEQ ID NO: 518.

74. The method of claim 65, wherein the anti-PD-1 antibody molecule comprises a heavy chain comprising the amino acid sequence of SEQ ID NO: 508 and a light chain comprising the amino acid sequence of SEQ ID NO: 522.

75. The method of claim 65, wherein the anti-PD-1 antibody molecule comprises a heavy chain comprising the amino acid sequence of SEQ ID NO: 508 and a light chain comprising the amino acid sequence of SEQ ID NO: 518.

76. The method of any one of claims 64-75, wherein the anti-PD-1 antibody molecule is administered to the subject at a dose of about 300 mg to 400 mg once every three weeks or once every four weeks (*e.g.*, about 400 mg once every four weeks).

77. The method of any one of claims 1-76, wherein the immuno-oncology therapy comprises a second immuno-oncology therapeutic agent (*e.g.*, an immuno-oncology therapeutic agent described herein).

78. The method of any one of claims 1-77, wherein the cancer is a solid tumor, a hematological cancer (*e.g.*, a leukemia, a lymphoma, or a myeloma), or a metastatic lesion thereof.

79. The method of claim 78, wherein the cancer is a melanoma or a metastatic lesion thereof.

80. The method of claim 79, wherein the melanoma is a stage I melanoma, a stage II melanoma, a stage III melanoma, or a stage IV melanoma.

81. The method of claim 78, wherein the cancer is a cancer other than a melanoma.

82. The method of claim 81, wherein the cancer is a lung cancer (*e.g.*, a non-small cell lung cancer), a pancreatic cancer, or a colorectal cancer, or a metastatic lesion thereof.

83. The method of any one of claims 78-82, wherein the cancer is refractory to an agent targeting BRAF and/or an agent targeting MEK.

84. The method of any one of claims 78-83, wherein the cancer (*e.g.*, melanoma) comprises a BRAF mutation.

85. The method of claim 84, wherein the BRAF mutation is a V600 mutation.

86. The method of claim 85, wherein the V600 mutation is a V600E or a V600K mutation.

87. The method of any one of claims 1-86, wherein the method further comprises administering an additional therapy to the subject (e.g., a pretreatment to the subject (e.g., to increase the value for the level or activity of CD4+ immune effector cells (*e.g.*, CD4+ T cells) relative to the level or activity of CD8+ immune effector cells (*e.g.*, CD8+ T cells) in the subject).

88. The method of claim 87, wherein the therapy is a first-line, second-line, third-line, or a fourth-line or beyond treatment.

89. The method of claim 87 or 88, wherein the therapy is an adjuvant treatment.

90. The method of any one of claims 87-89, wherein the therapy is a neoadjuvant treatment.

91. The method of any one of claims 1-79, comprising acquiring a value for the level and/or activity of immune activation comprising TILs, PD-L1, CD8, IFNγ, or a T-cell inflamed gene expression signature, *e.g.*, as described herein.

92. A method of identifying a subject having a cancer who is likely to benefit from a therapy, comprising:

acquiring a value for the level or activity of CD4+ immune effector cells (*e.g.*, CD4+ T cells) relative to the level or activity of CD8+ immune effector cells (*e.g.*, CD8+ T cells) in the subject (*e.g.*, in a sample from the subject),

wherein a value that is less than a reference value identifies the subject as one who is likely to benefit from the therapy, and

wherein the therapy comprises a targeted therapy (*e.g.*, without an immuno-oncology therapy).

93. The method of claim 92, wherein the subject is not likely to have a substantially increased benefit from a therapy comprising the targeted therapy in combination with an immuno-oncology therapy.

94. A method of selecting a therapy for a subject having a cancer, the method comprising:

acquiring a value for the level or activity of CD4+ immune effector cells (*e.g.*, CD4+ T cells) relative to the level or activity of CD8+ immune effector cells (*e.g.*, CD8+ T cells) in the subject (*e.g.*, in a sample from the subject), and

if the value is less than a reference value, selecting a therapy comprising a targeted therapy (*e.g.*, without an immuno-oncology therapy) for the subject.

95. The method of any one of claims 92-94, further comprising administering (*e.g.*, initiating administering or continuing administering) an effective amount of the therapy to the subject.

96. The method of any one of claims 92-94, further comprising administering an altered dosing regimen of the therapy (*e.g.*, a dosing regimen with a higher dose and/or more frequent administration than a reference dosing regimen) to the subject.

97. The method of any one of claims 92-94, further comprising discontinuing administration of a different therapy to the subject.

98. The method of any one of claims 92-94, further comprising administering an additional therapy to the subject.

99. The method of any one of claims 92-98, further comprising administering a pretreatment to the subject, wherein the pretreatment increases the value for the level or activity of CD4+ immune effector cells (*e.g.*, CD4+ T cells) relative to the level or activity of CD8+ immune effector cells (*e.g.*, CD8+ T cells) in the subject (*e.g.*, in a sample from the subject).

100. A method of treating a subject having a cancer, comprising:

responsive to a value for the level or activity of CD4+ immune effector cells (*e.g.*, CD4+ T cells) relative to the level or activity of CD8+ immune effector cells (*e.g.*, CD8+ T cells) in the subject (*e.g.*, in a sample from the subject), that is less than a reference value,

administering (*e.g.*, initiating administering or continuing administering) an effective amount of a therapy comprising targeted therapy (*e.g.*, without an immuno-oncology therapy) to the subject,

thereby treating the subject having the cancer.

101. A method of treating a subject having a cancer, comprising:

acquiring a value for the level or activity of CD4+ immune effector cells (*e.g.*, CD4+ T cells) relative to the level or activity of CD8+ immune effector cells (*e.g.*, CD8+ T cells) in the subject (*e.g.*, in a sample from the subject),

if the value is less than reference value, administering (*e.g.*, initiating administering or continuing administering) an effective amount of a therapy comprising a targeted therapy (*e.g.*, without an immuno-oncology therapy) to the subject,

thereby treating the subject having the cancer.

102. A method of treating a subject having a cancer, comprising:

administering (*e.g.*, initiating administering or continuing administering) an effective amount of a therapy comprising a targeted therapy in combination with an immuno-oncology therapy to the subject,

responsive to a value for the level or activity of CD4+ immune effector cells (*e.g.*, CD4+ T cells) relative to the level or activity of CD8+ immune effector cells (*e.g.*, CD8+ T cells) in the subject (*e.g.*, in a sample from the subject), that is less than a reference value,

administering (*e.g.*, initiating administering or continuing administering) an effective amount of a targeted therapy (*e.g.*, without an immuno-oncology therapy) to the subject,

thereby treating the subject having the cancer.

103. A method of treating a subject having a cancer, comprising:

administering (*e.g.*, initiating administering or continuing administering) an effective amount of a therapy comprising a targeted therapy (*e.g.*, without an immuno-oncology therapy) to the subject,

wherein prior to the administration, a value for the level or activity of CD4+ immune effector cells (*e.g.*, CD4+ T cells) relative to the level or activity of CD8+ immune effector cells (*e.g.*, CD8+ T cells) in the subject (*e.g.*, in a sample from the subject), that is less than a reference value, has been determined,

thereby treating the subject having the cancer.

104. A method of treating a subject having a cancer, comprising:

administering (*e.g.*, initiating administering or continuing administering) an effective amount of a therapy comprising a targeted therapy (*e.g.*, without an immuno-oncology therapy) to the subject,

wherein the subject is characterized as having a value for the level or activity of CD4+ immune effector cells (*e.g.*, CD4+ T cells) relative to the level or activity of CD8+ immune effector cells (*e.g.*, CD8+ T cells) in the subject (*e.g.*, in a sample from the subject), that is less than a reference value, thereby treating the subject having the cancer.

105. The method of any one of claims 92-104, wherein the value for the level or activity of CD4+ immune effector cells (*e.g.*, CD4+ T cells) relative to the level or activity of CD8+ immune effector cells (*e.g.*, CD8+ T cells) comprises a ratio of the amount of CD4+ immune effector cells (*e.g.*, CD4+ T cells) to the amount of CD8+ immune effector cells (*e.g.*, CD8+ T cells), *e.g.*, as measured by an assay disclosed herein, *e.g.*, flow cytometry immunophenotyping.

106. The method of claim 105, wherein the value is less than about 3.3 (e.g., less than 3.34).

107. The method of claim 106, wherein the value is less than about 3, 2.5, 2, 1.5, 1, or 0.5.

108. The method of claim 105, wherein the value is less than about 2 (e.g., less than 2.01).

109. The method of claim 108, wherein the value is less than about 1.5, 1, or 0.5.

110. The method of any one of claims 92-109, wherein acquiring the value comprises determining the level or activity of CD4+ immune effector cells (*e.g.*, CD4+ T cells) relative to the level or activity of CD8+ immune effector cells (*e.g.*, CD8+ T cells) in the subject, *e.g.*, in a sample from the subject.

111. The method of claim 110, wherein the sample from the subject comprises a blood sample (*e.g.*, a peripheral blood sample, *e.g.*, comprising peripheral blood mononuclear cells (PBMCs)) or a tumor sample.

112. The method of claim 110 or 111, wherein the value is acquired before administration of the therapy is initiated (*e.g.*, is a baseline value).

113. The method of claim 110 or 111 wherein the value is acquired after administration of the therapy is initiated.

114. The method of claim 113, wherein the value is acquired 1, 2, 3, 4, 8, 10, 12, 20, 30, 40 weeks or more or 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 12, 14, 16, 18, 20 months, or more after administration of the therapy is initiated.

115. The method of any one of claims 92-114, further comprising acquiring a value for TMB in the subject (*e.g.*, in a sample from the subject).

116. The method of claim 115, wherein a decreased value for TMB, as compared to a reference value, further identifies the subject as one who is likely to benefit from the therapy (*e.g.*, without an immuno-oncology therapy).

117. The method of claim 115, wherein a decreased value for TMB, as compared to a reference value, further identifies the subject as one who is not likely to have a substantially increased benefit from a therapy comprising the targeted therapy in combination with an immuno-oncology therapy.

118. The method of claim 116 or 117, wherein the value for TMB is less than 10 mut/Mb, *e.g.*, less than 9, 8, 7, 6, 5, 4, 3, 2, or 1 mut/Mb, or less.

119. The method of any one of claims 92-118, further comprising acquiring a value for the level and/or activity of PD-L1 in the subject (*e.g.*, in a sample from the subject).

120. The method of claim 119, wherein a decreased value for the level and/or activity of PD-L1, *e.g.*, together with a decreased value for TMB, further identifies the subject as one who is likely to benefit from the therapy (*e.g.*, without an immuno-oncology therapy).

121. The method of claim 119, wherein a decreased value for the level and/or activity of PD-L1, *e.g.*, together with a decreased value for TMB, further identifies the subject as one who is not likely to have a substantially increased benefit from a therapy comprising the targeted therapy in combination with an immuno-oncology therapy.

122. The method of any one of claims 119-121, wherein the cancer has low or no detectable expression of PD-L1.

123. The method of any one of claims 92-122, wherein the subject has been treated, or is being treated, with a targeted therapy.

124. The method of any one of claims 92-122, wherein the subject has not been treated, or is not being treated, with a targeted therapy.

125. The method of any one of claims 92-122, wherein the subject has received, or is receiving, with an immuno-oncology therapy.

126. The method of any one of claims 92-122, wherein the subject has not received, or is not receiving, with an immuno-oncology therapy.

127. The method of any one of claims 92-122, wherein the subject has received, or is receiving, with a therapy comprising a targeted therapy in combination with an immuno-oncology therapy.

128. The method of any one of claims 92-122, wherein the subject has not received, or is not receiving, with a therapy comprising a targeted therapy in combination with an immuno-oncology therapy.

129. The method of any one of claims 92-122, wherein the subject has received, or is receiving, with a targeted therapy, and the cancer has relapsed.

130. The method of any one of claims 92-122, wherein the subject is, or has been identified as, a non-responder to a targeted therapy.

131. The method of any one of claims 92-122, wherein the subject is, or has been identified as, a partial responder to a targeted therapy.

132. The method of any one of claims 92-131, wherein the targeted therapy comprises an agent targeting BRAF and/or an agent targeting MEK.

133. The method of claim 132, wherein the targeted therapy comprises an agent targeting BRAF or an agent targeting MEK.

134. The method of claim 132 or 133, wherein the agent targeting BRAF is a BRAF inhibitor.

135. The method of any one of claims 132-134, wherein the agent targeting BRAF inhibits wild-type BRAF and/or BRAF having a V600 mutation (*e.g.*, a V600E mutation or a V600K mutation).

136. The method of any one of claims 132-135, wherein the agent targeting BRAF is dabrafenib, vemurafenib, encorafenib, ABM-1310, ARQ-736, ASN003, BGB-283, BGB-3245, CEP-32496, GDC-0879, LUT014, PLX4720, PLX8394, RO5212054, or a pharmaceutically acceptable salt thereof.

137. The method of claim 136, wherein the agent targeting BRAF is dabrafenib.

138. The method of claim 136, wherein the agent targeting BRAF is vemurafenib.

139. The method of any one of claims 132-138, wherein the agent targeting BRAF (*e.g.* dabrafenib) is administered (*e.g.*, orally) at a dose between 25 mg and 300 mg (*e.g.*, between 50 mg and 250 mg or between 100 mg and 200 mg, *e.g.*, 150 mg), *e.g.*, twice a day.

140. The method of any one of claims 132-139, wherein the agent targeting MEK is a MEK inhibitor.

141. The method of any one of claims 132-140, wherein the agent targeting MEK is trametinib, cobimetinib, binimetinib, mirdametinib, pimasertib, refametinib, selumetinib, AS703988, AZD 8330, BI 847325, BIX 02188, BIX 02189, CI-1040, CS3006, E6201, FCN-159, G-38963, GDC-0623, HL-085, PD 98059, RO4987655, RO5126766, SHR 7390, TAK-733, U0126, WX-554, or a pharmaceutically acceptable salt thereof.

142. The method of claim 141, wherein the agent targeting MEK is trametinib.

143. The method of claim 141, wherein the agent targeting MEK is cobimetinib.

144. The method of any one of claims 132-143, wherein the agent targeting MEK (*e.g.*, trametinib) is administered (*e.g.*, orally) at a dose between 0.1 mg and 5 mg (*e.g.*, between 0.5 mg and 4 mg or between 1 mg and 3 mg, *e.g.*, at a dose of 2 mg), *e.g.*, once a day.

145. The method of any one of claims 132-137, 139-142, and 144, wherein the agent targeting BRAF is dabrafenib and the agent targeting MEK is trametinib.

146. The method of any one of claims 132-136, 138-141, 143, and 144, wherein the agent targeting BRAF is vemurafenib and the agent targeting MEK is cobimetinib.

147. The method of any one of claims 92-146, wherein the immuno-oncology therapy comprises a PD-1 or PD-L1 binding antagonist.

148. The method of any one of claims 92-147, wherein the immuno-oncology therapy comprises an PD-1 inhibitor.

149. The method of any one of claims 92-148, wherein the immuno-oncology therapy comprises an anti-PD-1 antibody molecule.

150. The method of claim 149, wherein the anti-PD-1 antibody molecule comprises:
a heavy chain variable region (VH) comprising a VHCDR1 amino acid sequence of SEQ ID NO: 501, a VHCDR2 amino acid sequence of SEQ ID NO: 502, and a VHCDR3 amino acid sequence of SEQ ID NO: 503; and
a light chain variable region (VL) comprising a VLCDR1 amino acid sequence of SEQ ID NO: 510, a VLCDR2 amino acid sequence of SEQ ID NO: 511, and a VLCDR3 amino acid sequence of SEQ ID NO: 512, each disclosed in Table 1.

151. The method of claim 150, wherein the anti-PD-1 antibody molecule comprises a VH comprising the amino acid sequence of SEQ ID NO: 506, or an amino acid sequence at least 85%, 90%, 95%, or 99% identical or higher to SEQ ID NO: 506.

152. The method of claim 150, wherein the anti-PD-1 antibody molecule comprises a VL comprising the amino acid sequence of SEQ ID NO: 520, or an amino acid sequence at least 85%, 90%, 95%, or 99% identical or higher to SEQ ID NO: 520.

153. The method of claim 150, wherein the anti-PD-1 antibody molecule comprises a VL comprising the amino acid sequence of SEQ ID NO: 516, or an amino acid sequence at least 85%, 90%, 95%, or 99% identical or higher to SEQ ID NO: 516.

154. The method of claim 150, wherein the anti-PD-1 antibody molecule comprises a VH comprising the amino acid sequence of SEQ ID NO: 506 and a VL comprising the amino acid sequence of SEQ ID NO: 520.

155. The method of claim 150, wherein the anti-PD-1 antibody molecule comprises a VH comprising the amino acid sequence of SEQ ID NO: 506 and a VL comprising the amino acid sequence of SEQ ID NO: 516.

156. The method of claim 150, wherein the anti-PD-1 antibody molecule comprises a heavy chain comprising the amino acid sequence of SEQ ID NO: 508, or an amino acid sequence at least 85%, 90%, 95%, or 99% identical or higher to SEQ ID NO: 508.

157. The method of claim 150, wherein the anti-PD-1 antibody molecule comprises a light chain comprising the amino acid sequence of SEQ ID NO: 522, or an amino acid sequence at least 85%, 90%, 95%, or 99% identical or higher to SEQ ID NO: 522.

158. The method of claim 150, wherein the anti-PD-1 antibody molecule comprises a light chain comprising the amino acid sequence of SEQ ID NO: 518, or an amino acid sequence at least 85%, 90%, 95%, or 99% identical or higher to SEQ ID NO: 518.

159. The method of claim 150, wherein the anti-PD-1 antibody molecule comprises a heavy chain comprising the amino acid sequence of SEQ ID NO: 508 and a light chain comprising the amino acid sequence of SEQ ID NO: 522.

160. The method of claim 150, wherein the anti-PD-1 antibody molecule comprises a heavy chain comprising the amino acid sequence of SEQ ID NO: 508 and a light chain comprising the amino acid sequence of SEQ ID NO: 518.

161. The method of claim 150, wherein the anti-PD-1 antibody molecule is administered to the subject at a dose of about 300 mg to 400 mg once every three weeks or once every four weeks (*e.g.*, about 400 mg once every four weeks).

162. The method of any one of claims 92-161, wherein the immuno-oncology therapy comprises a second immuno-oncology therapeutic agent (*e.g.*, an immuno-oncology therapeutic agent described herein).

163. The method of any one of claims 92-162, wherein the cancer is a solid tumor, a hematological cancer (*e.g.*, a leukemia, a lymphoma, or a myeloma), or a metastatic lesion thereof.

164. The method of claim 163, wherein the cancer is a melanoma or a metastatic lesion thereof.

165. The method of claim 164, wherein the melanoma is a stage I melanoma, a stage II melanoma, a stage III melanoma, or a stage IV melanoma.

166. The method of claim 163, wherein the cancer is a cancer other than a melanoma.

167. The method of claim 166, wherein the cancer is a lung cancer (*e.g.*, a non-small cell lung cancer), a pancreatic cancer, or a colorectal cancer, or a metastatic lesion thereof.

168. The method of any one of claims 92-167, wherein the cancer is refractory to an agent targeting BRAF and/or an agent targeting MEK.

169. The method of claim any one of claims 92-168, wherein the cancer (*e.g.*, melanoma) comprises a BRAF mutation.

170. The method of claim 169, wherein the BRAF mutation is a V600 mutation.

171. The method of claim 170, wherein the V600 mutation is a V600E or a V600K mutation.

172. The method of any one of claims 92-171, wherein the method further comprises administering an additional therapy to the subject (e.g., a pretreatment to the subject (e.g., to increase the value for the level or activity of CD4+ immune effector cells (*e.g.*, CD4+ T cells) relative to the level or activity of CD8+ immune effector cells (*e.g.*, CD8+ T cells) in the subject).

173. The method of claim 172, wherein the therapy is a first-line, second-line, third-line, or a fourth-line or beyond treatment.

174. The method of claim 172 or 173, wherein the therapy is an adjuvant treatment.

175. The method of any one of claims 172-174, wherein the therapy is a neoadjuvant treatment.

176. The method of any one of claims 92-175, comprising acquiring a value for the level and/or activity of immune activation comprising TILs, PD-L1, CD8, IFNγ, or a T-cell inflamed gene expression signature, *e.g.*, as described herein.

177. A method of stratifying subjects having a cancer into a first group and a second group, comprising:

acquiring a value for the level or activity of CD4+ immune effector cells (*e.g.*, CD4+ T cells) relative to the level or activity of CD8+ immune effector cells (*e.g.*, CD8+ T cells) in the subject (*e.g.*, in a sample from the subject),

assigning a subject who has a value that is less than a reference value to the first group who is likely to benefit from a therapy comprising a targeted therapy (*e.g.*, without an immuno-oncology therapy), and

assigning a subject who has a value that is greater than or equal to a reference vale to the second group which is likely to benefit from a therapy comprising a targeted therapy in combination with an immuno-oncology therapy.

178. A method of stratifying subjects having a cancer into a first group and a second group for selecting a therapy, comprising:

acquiring a value for the level or activity of CD4+ immune effector cells (*e.g.*, CD4+ T cells) relative to the level or activity of CD8+ immune effector cells (*e.g.*, CD8+ T cells) in the subject (*e.g.*, in a sample from the subject),

wherein a value that is less than a reference value identifies a subject as a member of the first group which is likely to benefit from a therapy comprising a targeted therapy (*e.g.*, without an immuno-oncology therapy), and

wherein a value that is greater than or equal to a reference value identifies a subject as a member of the second group which is likely to benefit from a therapy comprising a targeted therapy in combination with an immunotherapy.

179. The method of claim 177 or 178, further comprising administering (*e.g.*, initiating administering or continuing administering) an effective amount of the therapy to the subject.

180. The method of claim 177 or 178, further comprising administering an altered dosing regimen of the therapy (*e.g.*, a dosing regimen with a higher dose and/or more frequent administration than a reference dosing regimen) to the subject.

181. The method of claim 177 or 178, further comprising discontinuing administration of a different therapy to the subject.

182. The method of claim 177 or 178, further comprising administering an additional therapy to the subject.

183. The method of any one of claims 177-182, further comprising administering a pretreatment to the subject, wherein the pretreatment increases the value for the level or activity of CD4+ immune effector cells (*e.g.*, CD4+ T cells) relative to the level or activity of CD8+ immune effector cells (*e.g.*, CD8+ T cells) in the subject (*e.g.*, in a sample from the subject).

184. A method of treating a subject having a cancer, comprising:

responsive to a value for the level or activity of CD4+ immune effector cells (*e.g.*, CD4+ T cells) relative to the level or activity of CD8+ immune effector cells (*e.g.*, CD8+ T cells) in the subject (*e.g.*, in a sample from the subject),

administering (*e.g.*, initiating administering or continuing administering) an effective amount of a therapy comprising a targeted therapy (*e.g.*, without an immuno-oncology therapy) to a subject having a value that is less than a reference value; or

administering (*e.g.*, initiating administering or continuing administering) an effective amount of a therapy comprising a targeted therapy in combination with an immuno-oncology therapy to a subject having a value that is greater than or equal to a reference value,

thereby treating the subject having the cancer.

185. A method of treating a subject having a cancer, comprising:

acquiring a value for the level or activity of CD4+ immune effector cells (*e.g.*, CD4+ T cells) relative to the level or activity of CD8+ immune effector cells (*e.g.*, CD8+ T cells) in the subject (*e.g.*, in a sample from the subject); and

administering (*e.g.*, initiating administering or continuing administering) an effective amount of a therapy comprising a targeted therapy (*e.g.*, without an immuno-oncology therapy) to a subject having a value that is less than a reference value; or

administering (*e.g.*, initiating administering or continuing administering) an effective amount of a therapy comprising a targeted therapy in combination with an immuno-oncology therapy to a subject having a value that is greater than or equal to a reference value,

thereby treating the subject having the cancer.

186. A method of treating a subject having a cancer, comprising:

administering (*e.g.*, initiating administering or continuing administering) an effective amount of a therapy comprising a targeted therapy (*e.g.*, without an immuno-oncology therapy) to the subject; responsive to a value for the level or activity of CD4+ immune effector cells (*e.g.*, CD4+ T cells) relative to the level or activity of CD8+ immune effector cells (*e.g.*, CD8+ T cells) in the subject (*e.g.*, in a sample from the subject),

administering (*e.g.*, initiating administering or continuing administering) a therapy comprising a targeted therapy (*e.g.*, without an immuno-oncology therapy) for a subject having a value that is less than a reference value, or

administering (*e.g.*, initiating administering or continuing administering) a therapy comprising a targeted therapy in combination with an immuno-oncology therapy for a subject having a value that is greater than or equal to a reference value,

thereby treating the subject having the cancer.

187. A method of treating a subject having a cancer, comprising administering to the subject an effective amount of a therapy comprising:

(a) a targeted therapy (*e.g.*, without an immuno-oncology therapy), wherein prior to the administration, a value for the level or activity of CD4+ immune effector cells (*e.g.*, CD4+ T cells) relative to the level or activity of CD8+ immune effector cells (*e.g.*, CD8+ T cells) in the subject (*e.g.*, in a sample from the subject), that is less than a reference value, has been determined; or

(b) a targeted therapy in combination with an immuno-oncology therapy, wherein prior to the administration, a value for the level or activity of CD4+ immune effector cells (*e.g.*, CD4+ T cells) relative to the level or activity of CD8+ immune effector cells (*e.g.*, CD8+ T cells) in the subject (*e.g.*, in a sample from the subject), that is greater than or equal to a reference value, has been determined,

thereby treating the subject having the cancer.

188. A method of treating a subject having a cancer, comprising administering to the subject an effective amount of a therapy comprising:

(a) a targeted therapy (*e.g.*, without an immuno-oncology therapy) to the subject, wherein the subject is, or has been, characterized as having a value for the level or activity of CD4+ immune effector cells (*e.g.*, CD4+ T cells) relative to the level or activity of CD8+ immune effector cells (*e.g.*, CD8+ T cells) in the subject (*e.g.*, in a sample from the subject), that is less than a reference value; or

(b) a targeted therapy in combination with an immuno-oncology therapy, wherein the subject is, or has been, characterized as having a value for the level or activity of CD4+ immune effector cells (*e.g.*, CD4+ T cells) relative to the level or activity of CD8+ immune effector cells (*e.g.*, CD8+ T cells) in the subject (*e.g.*, in a sample from the subject), that is greater than or equal to a reference value,

thereby treating the subject having the cancer.

12189. The method according to any one of claims 1-6 or the therapy for use in a method of treating cancer according to any one of claims 7-11The method of any one of claims 177-188, wherein the value for the level or activity of CD4+ immune effector cells (*e.g.*, CD4+ T cells) relative to the level or activity of CD8+ immune effector cells (*e.g.*, CD8+ T cells) comprises a ratio of the amount of CD4+ immune effector cells (*e.g.*, CD4+ T cells) to the amount of CD8+ immune effector cells (*e.g.*, CD8+ T cells), *e.g.*, as measured by an assay disclosed herein, *e.g.*, flow cytometry immunophenotyping; optionally wherein

(a) the value that is greater than or equal to the reference value is.

190. The method of claim 189, wherein the value is

(i) greater than or equal to 2 (e.g., 2.01)., optionally

191. The method of claim 190, wherein the value is greater than or equal to 2.5, 3, 3.5, 4, 4.5, 5, 5.5, 6, 6.5, 7, 7.5, 8, 8.5, 9, 9.5, or 10, or

192. The method of claim 189, wherein the value is (ii) greater than or equal to 3.3 (e.g., 3.34), optionally

193. The method of claim 192, wherein the value is greater than or equal to 3.5, 4, 4.5, 5, 5.5, 6, 6.5, 7, 7.5, 8, 8.5, 9, 9.5, or 10, or

(b) the value that is less than the reference value is

194. The method of claim 189, wherein the value is

(i) less than about 3.3 (e.g., less than about 3.34), optionally

195. The method of claim 194, wherein the value is less than about 3, 2.5, 2, 1.5, 1, or 0.5, or

196. The method of claim 189, wherein the value is

(ii) less than about 2 (e.g., less than about 2.01), optionally

197. The method of claim 196, wherein the value is less than about 1.5, 1, or 0.5.

13198. The method according to any one of claims 1-6 or the therapy for use in a method of treating cancer according to any one of claims 7-12The method of any one of claims 177-197, wherein acquiring the value comprises determining the level or activity of CD4+ immune effector cells (*e.g.*, CD4+ T cells) relative to the level or activity of CD8+ immune effector cells (*e.g.*, CD8+ T cells) in the subject, *e.g.*, in a sample from the subject,

optionally

199. The method of claim 198, wherein the sample from the subject comprises a blood sample (*e.g.*, a peripheral blood sample, *e.g.*, comprising peripheral blood mononuclear cells (PBMCs)) or a tumor sample; and/or

200. The method of claim 198 or 199, wherein the value is acquired:

before administration of the therapy is initiated (*e.g.*, is a baseline value), or

201. The method of claim 198 or 199 wherein the value is acquired

after administration of the therapy is initiated,

optionally -

202. The method of claim 201, wherein the value is acquired 1, 2, 3, 4, 8, 10, 12, 20, 30, 40 weeks or more, or 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 12, 14, 16, 18, 20 months, or more after administration of the therapy is initiated.

14203. The method according to any one of claims 1-6 or the therapy for use in a method of treating cancer according to any one of claims 7-13The method of any one of claims 177-202, wherein the method further comprisesing acquiring a value for:

(a) the level or activity of CD8+ tumor infiltrating lymphocytes (TILs), *e.g.*, tumors having CD8+ TILs inflamed phenotype, in the subject (*e.g.*, in a sample from the subject), optionally:

204. The method of claim 203, wherein an increase in the value for the level or activity of CD8+ TILs, as compared to a reference value, further identifies the subject as one who is likely to benefit from the therapy; and/or

205. The method of any one of claims 177-204, further comprising acquiring a value for

(b) tumor mutation burden (TMB), in the subject, (*e.g.*, in a sample from the subject), optionally:

206. The method of claim 205, wherein:

an increased value for TMB, as compared to a reference value, further identifies the subject as one who is likely to benefit from the therapy, optionally:

207. The method of claim 206, wherein the value for TMB is greater than or equal to 10 mut/Mb, *e.g.*, greater than or equal to 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25 mut/Mb, or more, or

a decreased value for TMB, as compared to a reference value, further identifies the subject as one who is:

likely to benefit from the therapy (*e.g.*, without an immuno-oncology therapy), or

not likely to have a substantially increased benefit from a therapy comprising the targeted therapy in combination with an immuno-oncology

therapy; and/or wherein the value for TMB is less than 10 mut/Mb, *e.g.*, less than 9, 8, 7, 6, 5, 4, 3, 2, or 1 mut/Mb, or less; and/or

208. The method of any one of claims 177-207, further comprising acquiring a value for(c) the level and/or activity of PD-L1 in the subject (*e.g.*, in a sample from the subject),

optionally,

209. The method of claim 208, wherein:

a decreased value for the level and/or activity of PD-L1, as compared to a reference value, *e.g.*, together with an increased value for TMB, as compared to a reference value, further identifies the subject as one who is likely to benefit from the therapy; or

210. The method of claim 209, wherein the cancer has low or no detectable expression of PD-L1a decreased value for the level and/or activity of PD-L1, *e.g.*, together with a decreased value for TMB, further identifies the subject as one who is likely to benefit from the therapy (*e.g.*, without an immuno-oncology therapy); and/or.

211. The method of any one of claims 177-210 further comprising acquiring a value for

the cancer has low or no detectable expression of PD-L1.

(d) circulating tumor DNA (ctDNA) in the subject (*e.g.*, in a sample from the subject),

optionally

212. The method of claim 211, wherein an increased value for ctDNA, as compared to a reference value, further identifies the subject as one who is likely to benefit from the therapy.

213. The method of claim 205, wherein a decreased value for TMB, as compared to a reference value, further identifies the subject as one who is likely to benefit from the therapy (*e.g.*, without an immuno-oncology therapy).

214. The method of claim 205, wherein a decreased value for TMB, as compared to a reference value, further identifies the subject as one who is not likely to have a substantially increased benefit from a therapy comprising the targeted therapy in combination with an immuno-oncology therapy.

215. The method of claim 213 or 214, wherein the value for TMB is less than 10 mut/Mb, *e.g.*, less than 9, 8, 7, 6, 5, 4, 3, 2, or 1 mut/Mb, or less.

216. The method of any one of claims 213-215, further comprising acquiring a value for the level and/or activity of PD-L1 in the subject (*e.g.*, in a sample from the subject).

217. The method of claim 216, wherein a decreased value for the level and/or activity of PD-L1, *e.g.*, together with a decreased value for TMB, further identifies the subject as one who is likely to benefit from the therapy (*e.g.*, without an immuno-oncology therapy).

218. The method of claim 216, wherein a decreased value for the level and/or activity of PD-L1, *e.g.*, together with a decreased value for TMB, further identifies the subject as one who is not likely to have a substantially increased benefit from a therapy comprising the targeted therapy in combination with an immuno-oncology therapy.

219. The method of any one of claims 213-218, wherein the cancer has low or no detectable expression of PD-L1.

15220. The method according to any one of claims 1-6 or the therapy for use in a method of treating cancer according to any one of claims 7-14The method of any one of claims 177-219, wherein thea subject who is likely to benefit from, or is likely to have an increased benefit from, the therapy has an improved progression-free survival (PFS), duration of objective response (DOR), and/or overall survival (OS), compared to a subject who:

_____is unlikely to benefit from the therapy, or is unlikely to have an increased benefit from the therapy, or.

221. The method of any one of claims 177-219, wherein a subject who is likely to benefit from, or is likely to have an increased benefit from, the therapy has an improved PFS, DOR, and/or OS, compared to a subject who

has not received the therapy, or has only received targeted therapy but not an immuno-oncology therapy;

optionally,

222. The method of claim 220 or 221, wherein the PFS, DOR, and/or OS is improved by 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 18, 24, 30, 36, 42, 48, 54, 60 months or more, or

223. The method of claim 220 or 221, wherein the PFS, DOR, and/or OS is improved by 10%, 20%, 30%, 40%, 50%, 60%, 70%, 80%, or 90%, or 1, 2, 3, 4, 5, 6, 7, 8, 9, 10-fold or more.

16224. The method according to any one of claims 1-6 or the therapy for use in a method of treating cancer according to any one of claims 7-15The method of any one of claims 177-223, wherein the subject:

has been treated, or is being treated, with a targeted therapy;

225. The method of any one of claims 177-223, wherein the subject

has not been treated, or is not being treated, with a targeted therapy;

226. The method of any one of claims 177-223, wherein the subject

has received, or is receiving, with an immuno-oncology therapy;

227. The method of any one of claims 177-223, wherein the subject

has not received, or is not receiving, with an immuno-oncology therapy;

228. The method of claim any one of claims 177-223, wherein the subject

has received, or is receiving, with a therapy comprising a targeted therapy in combination with an immuno-oncology therapy;

229. The method of claim any one of claims 177-223, wherein the subject

has not received, or is not receiving, with a therapy comprising a targeted therapy in combination with an immuno-oncology therapy;

230. The method of any one of claims 177-223, wherein the subject

has received, or is receiving, with a targeted therapy, and the cancer has relapsed;

231. The method of any one of claims 177-223, wherein the subject

is, or has been identified as, a non-responder to a targeted therapy; or

232. The method of any one of claims 177-223, wherein the subject is, or has been identified as, a partial responder to a targeted therapy.

17233. The method according to any one of claims 1-6 or the therapy for use in a method of treating cancer according to any one of claims 7-16The method of any one of claims 177-232, wherein the targeted therapy comprises an agent targeting BRAF and/or an agent targeting MEK, optionally.

234. The method of claim 233, wherein:

the targeted therapy comprises an agent targeting BRAF or an agent targeting MEK; and/or.

235. The method of claim 233 or 234,

wherein the agent targeting BRAF is a BRAF inhibitor; and/or.

236. The method of any one of claims 233-235,

wherein the agent targeting BRAF inhibits wild-type BRAF and/or BRAF having a V600 mutation (*e.g.*, a V600E mutation or a V600K mutation); and/or.

237. The method of any one of claims 233-236,

wherein the agent targeting BRAF is dabrafenib, vemurafenib, encorafenib, ABM-1310, ARQ 736, ASN003, BGB-283, BGB-3245, CEP-32496, GDC-0879, LUT014, PLX4720, PLX8394, RO5212054, or a pharmaceutically acceptable salt thereof,

optionally .

238. The method of claim 237, wherein the agent targeting BRAF is dabrafenib or.

239. The method of claim 237, wherein the agent targeting BRAF is vemurafenib; and/or.

240. The method of claim any one of claims 232-239, wherein

the agent targeting BRAF (*e.g.* dabrafenib) is administered (*e.g.*, orally) at a dose between 25 mg and 300 mg (*e.g.*, between 50 mg and 250 mg or between 100 mg and 200 mg, *e.g.*, 150 mg), *e.g.*, twice a day; and/or.

241. The method of any one of claims 232-240, wherein

the agent targeting MEK is a MEK inhibitor; and/or.

242. The method of any one of claims 232-241, wherein

the agent targeting MEK is trametinib, cobimetinib, binimetinib, mirdametinib, pimasertib, refametinib, selumetinib, AS703988, AZD 8330, BI 847325, BIX 02188, BIX 02189, CI-1040, CS3006, E6201, FCN-159, G-38963, GDC-0623, HL-085, PD 98059, RO4987655, RO5126766, SHR 7390, TAK-733, U0126, WX-554, or a pharmaceutically acceptable salt thereof,

optionally ,

243. The method of claim 242, wherein the agent targeting MEK is trametinib or ,

244. The method of claim 242, wherein the agent targeting MEK is cobimetinib; and/or,

245. The method of any one of claims 232-244, wherein

the agent targeting MEK (*e.g.*, trametinib) is administered (*e.g.*, orally) at a dose between 0.1 mg and 5 mg (*e.g.,* between 0.5 mg and 4 mg or between 1 mg and 3 mg, *e.g.,* at a dose of 2 mg), *e.g.,* once a day; and/or,

246. The method of any one of claims 232-238, 240-243, or 245, wherein

the agent targeting BRAF is dabrafenib and the agent targeting MEK is trametinib, or ,

247. The method of any one of claims 232-237, 239-242, 244, or 245, wherein the agent targeting BRAF is vemurafenib and the agent targeting MEK is cobimetinib.

18248. The method according to any one of claims 1-6 or the therapy for use in a method of treating cancer according to any one of claims 7-17The method of any one of claims 177-247, wherein the immuno-oncology therapy comprises a PD-1 or PD-L1 binding antagonist; and/or,

249. The method of any one of claims 177-248, wherein the immuno-oncology therapy comprises an PD-1 inhibitor; and/or,

250. The method of any one of claims 177-249, wherein the immuno-oncology therapy comprises an anti-PD-1 antibody molecule; optionally wherein,

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251. The method of claim 250, wherein the anti-PD-1 antibody molecule comprises:

a heavy chain variable region (VH) comprising a VHCDR1 amino acid sequence of SEQ ID NO: 501, a VHCDR2 amino acid sequence of SEQ ID NO: 502, and a VHCDR3 amino acid sequence of SEQ ID NO: 503,; and

a light chain variable region (VL) comprising a VLCDR1 amino acid sequence of SEQ ID NO: 510, a VLCDR2 amino acid sequence of SEQ ID NO: 511, and a VLCDR3 amino acid sequence of SEQ ID NO: 512, each disclosed in Table 1;

optionally ;

252. The method of claim 251, wherein the anti-PD-1 antibody molecule comprises

a VH comprising the amino acid sequence of SEQ ID NO: 506, or an amino acid sequence at least 85%, 90%, 95%, or 99% identical or higher to SEQ ID NO: 506, ;

253. The method of claim 251, wherein the anti-PD-1 antibody molecule comprises a VL comprising the amino acid sequence of SEQ ID NO: 520, or an amino acid sequence at least 85%, 90%, 95%, or 99% identical or higher to SEQ ID NO: 520, ;

254. The method of claim 251, wherein the anti-PD-1 antibody molecule comprises a VL comprising the amino acid sequence of SEQ ID NO: 516, or an amino acid sequence at least 85%, 90%, 95%, or 99% identical or higher to SEQ ID NO: 516, ;

255. The method of claim 251, wherein the anti-PD-1 antibody molecule comprises a VH comprising the amino acid sequence of SEQ ID NO: 506 and a VL comprising the amino acid sequence of SEQ ID NO: 520;

256. The method of claim 251, wherein the anti-PD-1 antibody molecule comprises,

a VH comprising the amino acid sequence of SEQ ID NO: 506 and a VL comprising the amino acid sequence of SEQ ID NO: 516, ;

257. The method of claim 251, wherein the anti-PD-1 antibody molecule comprises a heavy chain comprising the amino acid sequence of SEQ ID NO:

508, or an amino acid sequence at least 85%, 90%, 95%, or 99% identical or higher to SEQ ID NO: 508.

258. The method of claim 251, wherein the anti-PD-1 antibody molecule comprises a light chain comprising the amino acid sequence of SEQ ID NO: 522, or an amino acid sequence at least 85%, 90%, 95%, or 99% identical or higher to SEQ ID NO: 522,

259. The method of claim 251, wherein the anti-PD-1 antibody molecule comprises a light chain comprising the amino acid sequence of SEQ ID NO: 518, or an amino acid sequence at least 85%, 90%, 95%, or 99% identical or higher to SEQ ID NO: 518,

260. The method of claim 251, wherein the anti-PD-1 antibody molecule comprises a heavy chain comprising the amino acid sequence of SEQ ID NO: 508 and a light chain comprising the amino acid sequence of SEQ ID NO: 522, or

261. The method of claim 251, wherein the anti-PD-1 antibody molecule comprises a heavy chain comprising the amino acid sequence of SEQ ID NO: 508 and a light chain comprising the amino acid sequence of SEQ ID NO: 518;

262. The method of any one of claims 250-261, optionally wherein wherein the anti-PD-1 antibody molecule is administered to the subject at a dose of about 300 mg to 400 mg once every three weeks or once every four weeks (*e.g.*, about 400 mg once every four weeks).

19263. The method according to any one of claims 1-6 or the therapy for use in a method of treating cancer according to any one of claims 7-18The method of any one of claims 177-262, wherein the immuno-oncology therapy comprises a second immuno-oncology therapeutic agent, (*e.g.*, an immuno-oncology therapeutic agent described herein).

20264. The method according to any one of claims 1-6 or the therapy for use in a method of treating cancer according to any one of claims 7-19The method of any one of claims 177-263, wherein the cancer is:

is a solid tumor, a hematological cancer (*e.g.*, a leukemia, a lymphoma, or a myeloma), or a metastatic lesion thereof,

optionally,

265. The method of claim 264, wherein the cancer:

is a melanoma or a metastatic lesion thereof,

optionally:

266. The method of claim 265, wherein the melanoma is a stage I melanoma, a stage II melanoma, a stage III melanoma, or a stage IV melanoma; or

267. The method of claim 264, wherein the cancer is a cancer other than a melanoma,

optionally,

268. The method of claim 267, wherein the cancer is a lung cancer (*e.g.*, a non-small cell lung cancer), a pancreatic cancer, or a colorectal cancer, or a metastatic lesion thereof; and/or

is

269. The method of any one of claims 177-268, wherein the cancer is refractory to an agent targeting BRAF and/or an agent targeting MEK; and/or

270. The method of claim any one of claims 177-269, wherein the cancer (*e.g.*, melanoma) comprises a BRAF mutation,

optionally

271. The method of claim 270, wherein the BRAF mutation is a V600 mutation, optionally:

272. The method of claim 271, wherein the V600 mutation is a V600E or a V600K mutation.

21~~273~~. The method according to any one of claims 1-6 or the therapy for use in a method of treating cancer according to any one of claims 7-20~~The method of any one of claims 177-272~~, wherein the method further comprises administering an additional therapy to the subject (e.g., a pretreatment to the subject (e.g., to increase the value for the level or activity of CD4+ immune effector cells (*e.g.*, CD4+ T cells) relative to the level or activity of CD8+ immune effector cells (*e.g.*, CD8+ T cells) in the subject);

optionally.

~~274. The method of claim 273,~~ wherein the therapy is

a first-line, second-line, third-line, or a fourth-line or beyond treatment, ~~and/or~~

~~275. The method of claim 273 or 274, wherein the therapy is~~ an adjuvant treatment, and/or.

~~276. The method of any one of claims 273-275, wherein the therapy is~~ a neoadjuvant treatment.

22~~277~~. The method according to any one of claims 1-6 or the therapy for use in a method of treating cancer according to any one of claims 7-21~~The method of any one of claims 177-276~~, wherein the method comprise~~s~~~~ing~~ acquiring a value for the level and/or activity of immune activation comprising TILs, PD-L1, CD8, IFNγ, or a T-cell inflamed gene expression signature, *e.g.*, as described herein.

~~278. A therapy comprising a targeted therapy in combination with an immuno-oncology therapy for use in a method of treating a subject having a cancer, comprising:~~

~~responsive to a value for the level or activity of CD4+ immune effector cells (*e.g.*, CD4+ T cells) relative to the level or activity of CD8+ immune effector cells (*e.g.*, CD8+ T cells) in the subject (*e.g.*, in a sample from the subject), that is greater than or equal to a reference value,~~

~~administering (*e.g.*, initiating administering or continuing administering) an effective amount of the therapy comprising the targeted therapy in combination with the immuno-oncology therapy to the subject.~~

~~279. A therapy comprising a targeted therapy in combination with an immuno-oncology therapy for use in a method of treating a subject having a cancer, comprising:~~

acquiring a value for the level or activity of CD4+ immune effector cells (*e.g.*, CD4+ T cells) relative to the level or activity of CD8+ immune effector cells (*e.g.*, CD8+ T cells) in the subject (*e.g.*, in a sample from the subject),

if the value is greater than or equal to a reference value, administering (*e.g.*, initiating administering or continuing administering) an effective amount of the therapy comprising the targeted therapy in combination with the immuno-oncology therapy to the subject.

280. A therapy comprising a targeted therapy in combination with an immuno-oncology therapy for use in a method of treating a subject having a cancer, comprising:

administering (*e.g.*, initiating administering or continuing administering) an effective amount of the targeted therapy to the subject;

responsive to a value for the level or activity of CD4+ immune effector cells (*e.g.*, CD4+ T cells) relative to the level or activity of CD8+ immune effector cells (*e.g.*, CD8+ T cells) in the subject (*e.g.*, in a sample from the subject), that is greater than or equal to a reference value,

administering (*e.g.*, initiating administering or continuing administering) an effective amount of the immuno-oncology therapy to the subject.

281. A therapy comprising a targeted therapy in combination with an immuno-oncology therapy for use in a method of treating a subject having a cancer, comprising:

administering (*e.g.*, initiating administering or continuing administering) an effective amount of the therapy comprising the targeted therapy in combination with the immuno-oncology therapy to the subject,

wherein prior to the administration, a value for the level or activity of CD4+ immune effector cells (*e.g.*, CD4+ T cells) relative to the level or activity of CD8+ immune effector cells (*e.g.*, CD8+ T cells) in the subject (*e.g.*, in a sample from the subject), that is greater than or equal to a reference value, has been determined.

282. A therapy comprising a targeted therapy in combination with an immuno-oncology therapy for use in a method of treating a subject having a cancer, comprising:

administering (*e.g.*, initiating administering or continuing administering) an effective amount of the therapy comprising the targeted therapy in combination with the immuno-oncology therapy to the subject,

wherein the subject is characterized as having a value for the level or activity of CD4+ immune effector cells (*e.g.*, CD4+ T cells) relative to the level or activity of CD8+ immune effector cells (*e.g.*, CD8+ T cells) in the subject (*e.g.*, in a sample from the subject), that is greater than or equal to a reference value.

283. A therapy comprising a targeted therapy for use in a method of treating a subject having a cancer, comprising:

responsive to a value for the level or activity of CD4+ immune effector cells (*e.g.*, CD4+ T cells) relative to the level or activity of CD8+ immune effector cells (*e.g.*, CD8+ T cells) in the subject (*e.g.*, in a sample from the subject), that is less than a reference value,

administering (*e.g.*, initiating administering or continuing administering) an effective amount of the therapy comprising targeted therapy (*e.g.*, without an immuno-oncology therapy) to the subject.

284. A therapy comprising a targeted therapy for use in a method of treating a subject having a cancer, comprising:

acquiring a value for the level or activity of CD4+ immune effector cells (*e.g.*, CD4+ T cells) relative to the level or activity of CD8+ immune effector cells (*e.g.*, CD8+ T cells) in the subject (*e.g.*, in a sample from the subject),

if the value is less than reference value, administering (*e.g.*, initiating administering or continuing administering) an effective amount of the therapy comprising the targeted therapy (*e.g.*, without an immuno-oncology therapy) to the subject.

285. A therapy comprising a targeted therapy for use in a method of treating a subject having a cancer, comprising:

administering (*e.g.*, initiating administering or continuing administering) an effective amount of the therapy comprising the targeted therapy in combination with an immuno-oncology therapy to the subject;

responsive to a value for the level or activity of CD4+ immune effector cells (*e.g.*, CD4+ T cells) relative to the level or activity of CD8+ immune effector cells (*e.g.*, CD8+ T cells) in the subject (*e.g.*, in a sample from the subject), that is less than a reference value,

administering (*e.g.*, initiating administering or continuing administering) an effective amount of the targeted therapy (*e.g.*, without an immuno-oncology therapy) to the subject.

286. A therapy comprising a targeted therapy for use in a method of treating a subject having a cancer, comprising:

administering (*e.g.*, initiating administering or continuing administering) an effective amount of the therapy comprising the targeted therapy (*e.g.*, without an immuno-oncology therapy) to the subject,

wherein prior to the administration, a value for the level or activity of CD4+ immune effector cells (*e.g.*, CD4+ T cells) relative to the level or activity of CD8+ immune effector cells (*e.g.*, CD8+ T cells) in the subject (*e.g.*, in a sample from the subject), that is less than a reference value, has been determined.

287. A therapy comprising a targeted therapy for use in a method of treating a subject having a cancer, comprising:

administering (*e.g.*, initiating administering or continuing administering) an effective amount of the therapy comprising the targeted therapy (*e.g.*, without an immuno-oncology therapy) to the subject,

wherein the subject is characterized as having a value for the level or activity of CD4+ immune effector cells (*e.g.*, CD4+ T cells) relative to the level or activity of CD8+ immune effector cells (*e.g.*, CD8+ T cells) in the subject (*e.g.*, in a sample from the subject), that is less than a reference value.

288. A therapy comprising a targeted therapy or comprising a targeted therapy in combination with an immuno-oncology therapy for use in a method of treating a subject having a cancer, comprising:

responsive to a value for the level or activity of CD4+ immune effector cells (*e.g.*, CD4+ T cells) relative to the level or activity of CD8+ immune effector cells (*e.g.*, CD8+ T cells) in the subject (*e.g.*, in a sample from the subject),

administering (*e.g.*, initiating administering or continuing administering) an effective amount of the therapy comprising the targeted therapy (*e.g.*, without an immuno-oncology therapy) to a subject having a value that is less than a reference value; or

administering (*e.g.*, initiating administering or continuing administering) an effective amount of the therapy comprising the targeted therapy in combination with the immuno-oncology therapy to a subject having a value that is greater than or equal to a reference value.

289. A therapy comprising a targeted therapy or comprising a targeted therapy in combination with an immuno-oncology therapy for use in a method of treating a subject having a cancer, comprising:

acquiring a value for the level or activity of CD4+ immune effector cells (*e.g.*, CD4+ T cells) relative to the level or activity of CD8+ immune effector cells (*e.g.*, CD8+ T cells) in the subject (*e.g.*, in a sample from the subject); and

administering (*e.g.*, initiating administering or continuing administering) an effective amount of the therapy comprising the targeted therapy (*e.g.*, without an immuno-oncology therapy) to a subject having a value that is less than a reference value; or

administering (*e.g.*, initiating administering or continuing administering) an effective amount of the therapy comprising the targeted therapy in combination with the immuno-oncology therapy to a subject having a value that is greater than or equal to a reference value.

290. A therapy comprising a targeted therapy or comprising a targeted therapy in combination with an immuno-oncology therapy for use in a method of treating a subject having a cancer, comprising:

administering (*e.g.*, initiating administering or continuing administering) an effective amount of the therapy comprising the targeted therapy (*e.g.*, without an immuno-oncology therapy) to the subject;

responsive to a value for the level or activity of CD4+ immune effector cells (*e.g.*, CD4+ T cells) relative to the level or activity of CD8+ immune effector cells (*e.g.*, CD8+ T cells) in the subject (*e.g.*, in a sample from the subject),

administering (*e.g.*, initiating administering or continuing administering) the therapy comprising the targeted therapy (*e.g.*, without an immuno-oncology therapy) for a subject having a value that is less than a reference value, or

administering (*e.g.*, initiating administering or continuing administering) the therapy comprising the targeted therapy in combination with the immuno-oncology therapy for a subject having a value that is greater than or equal to a reference value.

291. A therapy comprising a targeted therapy or comprising a targeted therapy in combination with an immuno-oncology therapy for use in a method of treating a subject having a cancer, comprising administering to the subject an effective amount of the therapy comprising:

(a) the targeted therapy (*e.g.*, without an immuno-oncology therapy), wherein prior to the administration, a value for the level or activity of CD4+ immune effector cells (*e.g.*, CD4+ T cells) relative to the level or activity of CD8+ immune effector cells (*e.g.*, CD8+ T cells) in the subject (*e.g.*, in a sample from the subject), that is less than a reference value, has been determined; or

(b) the targeted therapy in combination with the immuno-oncology therapy, wherein prior to the administration, a value for the level or activity of CD4+ immune effector cells (*e.g.*, CD4+ T cells) relative to the level or activity of CD8+ immune effector cells (*e.g.*, CD8+ T cells) in the subject (*e.g.*, in a sample from the subject), that is greater than or equal to a reference value, has been determined.

292. A therapy comprising a targeted therapy or comprising a targeted therapy in combination with an immuno-oncology therapy for use in a method of treating a subject having a cancer, comprising administering to the subject an effective amount of the therapy comprising:

(a) the targeted therapy (*e.g.*, without an immuno-oncology therapy) to the subject, wherein the subject is, or has been, characterized as having a value for the level or activity of CD4+ immune effector cells (*e.g.*, CD4+ T cells) relative to the level or activity of CD8+ immune effector cells (*e.g.*, CD8+ T cells) in the subject (*e.g.*, in a sample from the subject), that is less than a reference value; or

(b) the targeted therapy in combination with the immuno-oncology therapy, wherein the subject is, or has been, characterized as having a value for the level or activity of CD4+ immune effector cells (*e.g.*, CD4+ T cells) relative to the level or activity of CD8+ immune effector cells (*e.g.*,

~~CD8+ T cells) in the subject (*e.g.*, in a sample from the subject), that is greater than or equal to a reference value.~~